

ARCHIMEDES Preparing BIDS Data for Submission

A guide for research teams sharing MRI and EEG data in BIDS format with ARCHIMEDES

This guide explains what to send when sharing imaging or EEG data organised in BIDS (Brain Imaging Data Structure). If anything here does not fit your study, contact the ARCHIMEDES technical support team before sending. We are happy to help.

You only need to provide one thing

Your **BIDS dataset**, including a **participants.tsv** file with date of birth and sex for each participant. Removing identifying information from the images is done on our side.

1. Where to put your files

```
/project_name/
├── deidentified-raw/
│   └── bids/
│       ├── dataset_description.json
│       ├── participants.tsv
│       ├── sub-ARCHI0001/
│       │   ├── ses-V01/
│       │   │   ├── anat/
│       │   │   │   ├── sub-ARCHI0001_ses-V01_T1w.nii.gz
│       │   │   │   └── sub-ARCHI0001_ses-V01_T1w.json
│       │   │   └── eeg/
│       │   │       ├── sub-ARCHI0001_ses-V01_task-rest_eeg.edf
│       │   │       └── sub-ARCHI0001_ses-V01_task-rest_eeg.json
│       │   └── ses-V02/
│       │       └── anat/
│       └── sub-ARCHI0002/
│           ├── ses-V01/
│           └── anat/
```

Put everything inside one top folder named after your project, for example CNMDP/, and share that folder with us. The project name is how we identify which project the files belong to.

- One sub- folder per participant, and one ses- folder per visit inside it, even if your study has only one visit.
- Inside each visit, files go in the folder for their type, such as anat, func, dwi, fmap or eeg.
- Each file name starts with the same participant and visit labels as the folders it sits in.
- MRI and EEG for the same participants go in the same dataset.

Example BIDS datasets: <https://github.com/bids-standard/bids-examples>

2. participants.tsv

One row per participant, saved at the top of the bids/ folder.

Column	Needed	What to enter	Example
participant_id	Yes	The participant's folder name	sub-ARCHI0001
dob	Yes	Date of birth as YYYY-MM-DD, YYYY-MM or YYYY	1948-03
sex	Yes	Male, Female or Other	Male

Column	Needed	What to enter	Example
site	Preferred	Your site name. If not sure, check with the ARCHIMEDES technical support team, who can send you the correct name	University of Ottawa Heart Institute
cohort	Preferred	The participant's group. If not sure, check with the ARCHIMEDES technical support team, who can send you the correct name	Affected
project	Preferred	Your project name. If not sure, check with the ARCHIMEDES technical support team, who can send you the correct name	CNMDP

Example participants.tsv

	A	B	C	D	E	F
1	participant_id	dob	sex	site	cohort	project
2	sub-ARCHI0001	1948-03	Male	University of Ottawa Heart Institute	Affected	CNMDP
3	sub-ARCHI0002	1962-11	Female	University of Ottawa Heart Institute	Control	CNMDP
4	sub-ARCHI0003	1955	Other	University of Ottawa Heart Institute	Control	CNMDP

Build it in Excel or Google Sheets if that is easiest, then save it as **Tab Delimited Text** (Excel) or **Tab-separated values** (Sheets), named participants.tsv. Put the column names on the first line, with no title rows above them.

Only the year and month of birth are kept, so send whatever date precision you are comfortable with.

3. dataset_description.json

Every BIDS dataset needs this small file at the top of the bids/ folder. Most BIDS conversion tools create it for you. A minimal version looks like this:

```
{
  "Name": "CNMDP imaging",
  "BIDSVersion": "1.9.0"
}
```

4. Simple rules

Rule	Good	Avoid
Participant IDs use letters and numbers only	sub-ARCHI0001	sub-ARCHI_0001, sub-ARCHI-0001
All participant IDs follow the same pattern: same letters, same number of digits	sub-ARCHI0001, sub-ARCHI0002	sub-ARCHI0001, sub-P14
Same ID for a participant across all visits and data types	sub-ARCHI0001 in MRI and EEG	sub-ARCHI0001, then sub-ARCHI1
Visit names short, consistent and as agreed with us	ses-V01, ses-V02	ses-visit1, ses-V1 then ses-V01
Every participant folder has a row in participants.tsv	Folder and row for each	A folder with no row

Rule	Good	Avoid
Dates as YYYY-MM-DD, YYYY-MM or YYYY only	1948-03-22, 1948-03, 1948	Any other format
Sex written in full	Male, Female, Other	M, F, 1, 2
No names or identifying information in IDs or file names	sub-ARCHI0001	sub-SmithJ

5. Before you send

Check your dataset

Run the free BIDS Validator (<https://bids-standard.github.io/bids-validator/>) on your dataset. It runs in your browser and your files are not uploaded. Warnings are fine; errors need fixing, because a dataset with errors cannot be loaded.

Good to know

- Tell us your visit names before your first upload, so we can set them up.
- You do not need to remove identifying information from the image or EEG files. We do that on our side. Just keep names and other identifying information out of IDs, folder names, file names and participants.tsv.
- To send more data later, add the new participants or visits to the same dataset and add new participants to participants.tsv.
- To replace data that was already sent, contact the ARCHIMEDES technical support team.
- Questionnaire or clinical data should be sent separately, following the REDCap or tabular data guide.
- If something in your dataset cannot be loaded, we will contact you.

6. Common mistakes to avoid

What to avoid in participants.tsv (shaded)

	A	B	C
1	participant_id	dob	sex
2	sub-ARCHI0001	1948-03	Male
3	sub-ARCHI_0002	03/22/1962	F
4	sub-P14	1955	Other

Avoid	Example	Instead
A participant folder that is not listed in participants.tsv	sub-ARCHI0003/ with no row	A row for every folder
IDs in different patterns	sub-ARCHI0001, sub-P14	Same letters and number of digits
Underscores or dashes inside an ID	sub-ARCHI_0002	sub-ARCHI0002
File labels that do not match the folders	sub-ARCHI0001_ses-1_T1w.nii.gz in ses-V01/	sub-ARCHI0001_ses-V01_T1w.nii.gz

Avoid	Example	Instead
participants.tsv saved in another format	.csv or .xlsx	Tab-separated, named participants.tsv
Sex as a letter or code	F, M	Female, Male or Other
Date of birth in another format	03/22/1962	1962-03-22, 1962-03 or 1962
Project or cohort names that differ from the accepted list	Different spelling or extra spaces	Exactly as on the list we send you
Missing dataset_description.json	No file at the top of bids/	Add the file

7. Checklist

Files ☐ BIDS dataset in deidentified-raw/bids/ ☐ dataset_description.json and participants.tsv at the top ☐ A sub- folder per participant, a ses- folder per visit ☐ BIDS Validator run with no errors	participants.tsv ☐ A row for every participant folder ☐ dob and sex on every row ☐ site, cohort and project filled; if not sure, check with support ☐ IDs follow one pattern, letters and numbers only ☐ Saved as tab-separated
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